HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
EEE127: Electronic Devices and Circuits

Batch:29th | Semester: Spring 2026
Time: 1 Hour 30 Minutes

Date:08/06/2026 **Midterm Examination** Total Marks: 30

Answer any **five** questions. $5 \times 6 = 30$

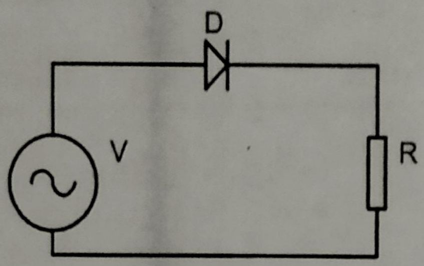
1.	(a)	Explain the band theory of solids.	CLO1	2
	(b)	Based on band theory, describe the electrical conductivity of different types of elements (conductors, semiconductors, insulators).	CLO1	4
2.	(a)	Define the following terms: i) Conduction band ii)Valence band iii)Forbidden energy gap	CLO1	3
	(b)	Can a trivalent element increase the conductivity of a semiconductor? Justify your answer.	CLO1	3
3.	(a)	What are the characteristics of an ideal diode?	CLO2	2
	(b)	Describe the V-I characteristics of a PN junction diode in detail, highlighting the behavior under forward and reverse bias conditions.	CLO2	4
4.	(a)	Describe the differences between majority and minority charge carriers of semiconductors.	CLO2	3
	(b)	A PN junction diode is connected with its negative terminal to the positive supply and its positive terminal to the negative supply. Identify the type of biasing and briefly describe its effect.	CLO1	3
5.	(a)	Define and describe the operation of a half-wave rectifier with appropriate circuit diagram.	CLO1	3
	(b)	Derive the expressions for the DC voltage and DC current of a half-wave rectifier.	CLO2	3

6. (a) Define the following terms: CLO1 1+1=2

i) Full wave rectifier ii) DC voltage of a rectifier

2+2=4

(b) Observe the following circuit diagram and answer the following questions: CLO2



$V = 150\sin\theta$

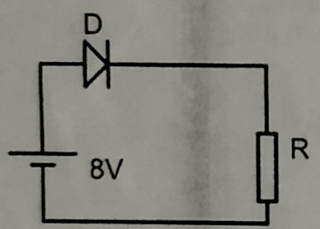
$R_d = 300\Omega$

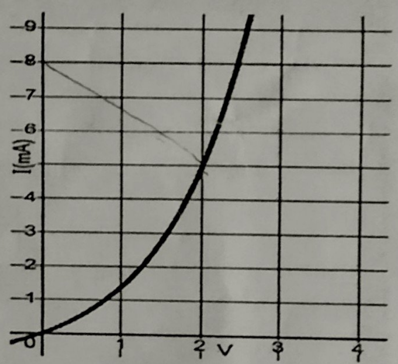
$R = 3k\Omega$

i) Calculate the dc output voltage of the rectifier.

ii) Calculate the efficiency of the given rectifier.

7. Observe the following circuit diagram and the V-I characteristics of the diode, then answer the following questions. CLO2 2+3+1=6





R=1kΩ

Figure: V-I characteristics of diode D

i) Find out the operating point.

ii) Calculate the voltage drop and power dissipated across the load resistor.

iii) Calculate the current flowing through the diode.

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 121 : Structured Programming Language

Batch: 29th | Semester: Spring 2025

Time: 1 Hour 30 Minutes

Date: 13/06/2026 | **Midterm Examination** | Total Marks: 30

Answer any five questions. $5 \times 6 = 30$

1. (a) Define the following: — CLO1 — $2\times1=2$
   i. Programming Language
   ii. Debugging

   (b) Write the algorithm and draw the corresponding flowchart to find even numbers between 1 and 50. — CLO2 — 4

2. (a) What is the role of `#include` directive in C programming? — CLO4 — 2

   (b) Find errors, if any, in the following programs — — CLO5 — $4\times1=4$

i.
```
#include <stdio.h>
int main() {
    int x;
    scanf("%d", x);
    printf("%d", x);
    return 0;
}
```

ii.
```
#include <stdio.h>
void main() {
    int n = 5;
    printf("%d\n", n)
    return 0;
}
```

iii.
```
#include <stdio.h>
int main() {
    int a = 10;
    int b = 20;
    sum = a + b;
    printf("%d", sum);
    return 0;
}
```

iv.
```
#include <stdio.h>
int main() {
    float num
    printf("%f", num);
    return 0;
}
```

3. (a) State the rules for naming an identifier in C. — CLO4 — 2

   (b) Classify the following variable names as valid or invalid. For the invalid ones state the reason. — CLO5 — $4\times.5=2$
   i. `_temp123`
   ii. `student@id`
   iii. `RollNo1`
   iv. `main`

   (c) Write a C program to swap the values of two integer variables and display the values after swapping. — CLO3 — 2

4. (a) Differentiate between implicit and explicit type conversions with examples. — CLO3 — 3

(b) Find out the output of the following code snippet — — CLO4 — 3

```
#include <stdio.h>
int main() {
    int x, y, z, w;
    x = 7;
    y = 11;
    z = ++x + y--;
    printf("x = %d y = %d z = %d\n", x, y, z);
    w = x-- * ++z;
    printf("x = %d z = %d w = %d\n", x, z, w);
    printf("w /= z = %d\n", w /= z);
    printf("%d\n", (x > y) ? x - y : y - x);
    return 0;
}
```

5. (a) Determine the output for the following `printf` statements: — CLO4 — 2

```
float num = 123.4567;
printf(“%f\n”, num);
printf(“%8.3f\n”, num);
printf(“%10.2f\n”, num);
printf(“%7.1f\n”, num);
```

(b) Write the following programs in C — — CLO4 — 2×2=4

i. Program to check if a given character is alphabet or digit.

ii. Program to read a character and convert it to the opposite case (uppercase ↔ lowercase).

6. (a) Write a C program that takes a numerical score (0–100) and displays the corresponding grade using a `switch-case` statement. The grading scheme is as follows: — CLO4 — 3

- 80–100 → A
- 70–79 → B
- 60–69 → C
- Below 60 → F

(b) Convert between different Loop Structures in C – — CLO4 — 2×1.5=3

i. Convert the following `while` loop into a `for` loop.

```
int i = 2, count = 0;
while(count < 5) {
    printf("%d ", i);
    i += 2;
    count++;
}
```

ii. Convert the following `for` loop into a `while` loop.

```
int n, sum = 0;
scanf("%d", &n);
for (int i = 1; i <= n; i++) {
    sum += i;
}
```

7. (a) Differentiate between Counter-controlled and Sentinel-controlled loops. — CLO3 — 2

(b) Write a C program to generate the Fibonacci series up to n terms. The value of n should be taken as input from the user. — CLO4 — 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science and Engineering
Department of Computer Science and Engineering
BSC (Honors) in CSE
ENG 127: English Skills
Mid-Term Examination
Spring 2026

Batch: 29th | 2nd Semester
June 15, 2026 | Time: 1 Hour 30 Minutes

Total Marks: 30

CLOs	**Question/ Statement**	**Level**	**Marks**
CLO1	**Read the passage carefully and answer questions 1-20 from the passage. Each question carries 01 mark.**	Understand	20×1=20

Roman Shipbuilding and Navigation

Shipbuilding today is based on science and ships are built using computers and sophisticated tools. Shipbuilding in ancient Rome, however, was more of an art relying on estimation, inherited techniques and personal experience. The Romans were not traditionally sailors but mostly land-based people, who learned to build ships from the people that they conquered, namely the Greeks and the Egyptians.

There are a few surviving written documents that give descriptions and representations of ancient Roman ships, including the sails and rigging. Excavated vessels also provide some clues about ancient shipbuilding techniques. Studies of these have taught us that ancient Roman shipbuilders built the outer hull first, then proceeded with the frame and the rest of the ship. Planks used to build the outer hull were initially sewn together. Starting from the 6th century BCE, they were fixed using a method called mortise and tenon, whereby one plank locked into another without the need for stitching. Then in the first centuries of the current era, Mediterranean shipbuilders shifted to another shipbuilding method, still in use today, which consisted of building the frame first and then proceeding with the hull and the other components of the ship. This method was more systematic and dramatically shortened ship construction times. The ancient Romans built large merchant ships and warships whose size and technologies were unequalled until the 16th century CE.

Warships were built to be lightweight and very speedy. They had to be able to sail near the coast, which is why they had no ballast or excess load and were built with a long, narrow hull. They did not sink when damaged and often would lie crippled on the sea's surface following naval battles. They had a bronze battering ram, which was used to pierce the timber hulls or break the oars of enemy vessels. Warships used both wind (sails) and human power (oarsmen) and were therefore very fast. Eventually, Rome's navy became the largest and most powerful in the Mediterranean, and the Romans had control over what they therefore called Mare Nostrum meaning 'our sea'.

There were many kinds of warship. The 'trireme' was the dominant warship from the 7th to 4th century BCE. It had rowers in the top, middle and lower levels, and approximately 50 rowers in each bank. The rowers at the bottom had the most uncomfortable position as they were under the other rowers and were exposed to the water entering through the oar-holes. It is worth noting that contrary to popular perception, rowers were not slaves but mostly Roman citizens enrolled in the military. The trireme was superseded by larger ships with even more rowers.

Merchant ships were built to transport lots of cargo over long distances and at a reasonable cost. They had a wider hull, double planking and a solid interior for added stability. Unlike warships, their V-shaped hull was deep underwater, meaning that they could not sail too close to the coast. They usually had two huge side rudders located off the stern and controlled by a small tiller bar connected to a system of cables. They had from one to three masts with large square sails and a small triangular sail at the bow. Just like warships, merchant ships used oarsmen, but coordinating the hundreds of rowers in both types of ship was not an easy task. In order to assist them, music would be played on an instrument, and oars would then keep time with this.

The cargo on merchant ships included raw materials (e.g. iron bars, copper, marble and granite), and agricultural products (e.g. grain from Egypt's Nile valley). During the Empire, Rome was a huge city by ancient standards of about one million inhabitants. Goods from all over the world would come to the city through the port of Pozzuoli situated west of the bay of Naples in Italy and through the gigantic port of Ostia situated at the mouth of the Tiber River. Large merchant ships would approach the destination port and, just like today, be intercepted by a number of towboats that would drag them to the quay.

The time of travel along the many sailing routes could vary widely. Navigation in ancient Rome did not rely on sophisticated instruments such as compasses but on experience, local knowledge and observation of natural phenomena. In conditions of good visibility, seamen in the Mediterranean often had the mainland or islands in sight, which greatly facilitated navigation. They sailed by noting their position relative to a succession of recognizable landmarks. When weather conditions were not good or where land was no longer visible, Roman mariners estimated directions from the pole star or, with less accuracy, from the Sun at noon. They also estimated directions relative to the wind and swell. Overall, shipping in ancient Roman times resembled shipping today with large vessels regularly crossing the seas and bringing supplies from their Empire.

CLO5 TRUE **if the statement agrees with the information** Analyze 7×1=7
FALSE **if the statement contradicts the information**
NOT GIVEN **if there is no information on this**

1. The Romans' shipbuilding skills were passed on to the Greeks and the Egyptians.
2. Skilled craftsmen were needed for the mortise and tenon method of fixing planks.
3. The later practice used by Mediterranean shipbuilders involved building the hull before the frame.
4. The Romans called the Mediterranean Sea Mare Nostrum because they dominated its use.
5. Most rowers on ships were people from the Roman army.
6. Large merchant ships carried goods such as grain and marble to Rome.
7. Roman sailors used compasses to navigate when land was not visible.

CLO2 **Fill the blanks**

Choose NO MORE THAN ONE WORD AND/OR A NUMBER from the passage for each answer. Understand 7×1=7

Warships and merchant ships

Warships were designed so that they were 8.. and moved quickly. They often remained afloat after battles and were able to sail close to land as they lacked any additional weight. A battering ram made of 9.. was included in the design for attacking and damaging the timber and oars of enemy ships. Warships, such as the 'trireme', had rowers on three different 10..

Unlike warships, merchant ships had a broad 11.. that lay far below the surface of the sea. Merchant ships were steered through the water with the help of large rudders and a tiller bar. They had both square and 12.. sails.

On merchant ships and warships, 13.. was used to ensure rowers moved their oars in and out of the water at the same time.

Quantities of agricultural goods such as 14.. were transported by merchant ships to two main ports in Italy.

CLO5 **Choose the correct letter A, B, C or D from the following questions.** Analyze 06×1=06

15. The passage says the Romans learned to build ships from ____.
A. the Persians B. the Greeks and the Egyptians
C. their own soldiers D. Chinese traders

16. The bronze battering ram on Roman warships was used to ____.
A. steer the ship B. carry extra cargo
C. damage enemy ships or break their oars D. signal other Roman ships

17. A 'trireme' was a type of ____.
A. large cargo ship B. Roman warship with rowers on three levels
C. navigation tool D. type of sail

18. Roman merchant ships could not sail too close to the coast because ____.
A. they were too wide to turn B. their hull went very deep underwater
C. their sails needed strong winds D. coastal ports were too small

19. When Roman sailors could not see land, they found direction by looking at ____.
A. a compass B. other ships C. the pole star or the Sun D. hand-drawn maps

20. On both warships and merchant ships, rowers were helped to keep time by ____.
A. the captain shouting orders B. hand signals from the front
C. music played on an instrument D. a mechanical device on deck

CLO5 **21. Write a Summary on the passage within 70 words** Understand 5×1=5

CLO1 **22. What is scanning in reading? Mention at least three examples of scanning.** Define 5×1=5

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT 125 **Course Title: Linear Algebra and Complex Variables**
Midterm Examination Semester: Spring 2026
Date: 20/06/2026 Total Time: 1:30 hours **Total Marks: 30**

Answer any three (03) out of the following questions: $3 \times 10 = 30$
[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.		Questions	CLOs	Marks
Q1.	(a)	Define a triangular matrix and an inverse matrix.	CLO1	[2]
	(b)	If $C=\begin{bmatrix}1&3&2\\2&0&3\\1&-1&1\end{bmatrix}$ then find $P(x)=2x^2-4x-2I$ and hence find $P(C^{-1})$	CLO3	[8]
Q2.	(a)	Define a vector space over a field. Give two examples: one over $\mathbb{R}$ *and* $\mathbb{C}$ one over	CLO1	[4]
	(b)	Which of the following sets of vectors $\mathbb{R}^3$ are linearly independent? (i) {(1, 0, 1), (-3, 2, 6), (4, 5, -2)} (ii) {(1, -4, 2), (3, -5, 1), (2, 7, 8), (-1, 1, 1)}	CLO3	[6]
Q3.	(a)	In digital signal processing, signals are represented as vectors in $\mathbb{R}^n$. A set of all even signals is defined as: $W=\{f:\mathbb{Z}\to\mathbb{R} \mid f(-1)=f(n) \text{ for all } n\}$. Show that **W** forms a subspace of the vector space of all real-valued signals. What is the physical significance of this subspace?	CLO1	[5]
	(b)	In 3D computer graphics, the set of all vectors lying on a plane through the origin is given by: $P=\{(x,y,z)\in\mathbb{R}^3 : x+2y-z=0\}$ Show P is a subspace of $\mathbb{R}^3$ and find its basis and dimension.	CLO3	[5]
Q4.	(a)	Let T: $\mathbb{R}^3\to\mathbb{R}^3$ be the linear operator defined by $T(x,y,z)=(x+2y, y-z, x+2z)$. Find the rank and nullity of T.	CLO3	[10]

Q5. A telemetry polynomial $p(t) = 2t^3 - 3t^2 - t + 5 \in P_3$ models the drone's altitude profile over time, and the differentiation operator $D: P_3 \rightarrow P_2$ gives the instantaneous vertical velocity. CLO3 [10]

(i) Using the standard bases $B_1 = \{1, t, t^2, t^3\}$ for P_3 and $B_2 = \{1, t, t^2\}$ for p_2, write dawn the matrix $[D]_{B_2}^{B_1}$ of the differentiation operator.

(ii) Compute $[D]_{B_2}^{B_1}\,[P]_{B_1}$ to find $[P'(t)]_{B_2}$, and hence write down p′(t) explicitly. Interpret p′(t) as the drone's instantaneous vertical velocity.

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY 121 | **Course Title: Physics II**
Midterm Examination | Semester: Spring 2026
Date: 6/22/2026 | Total Time: 1 Hour 30 Minutes | ***Total Marks: 30***

<u>Answer any three (03) out of the following questions</u>: $3 \times 10 = 30$

[The figures in the right margin indicate full marks. The symbols used have their usual meaning.]

			CLOs	**Marks**
Q1.	**(a)**	Define the terms: (i) Charge (ii) Conductor (iii) Insulator (iv) Point Charge.	CLO1	[2]
	(b)	Write down the properties of electric charge.	CLO2	[2]
	(c)	State and explain Coulomb's law of electrostatics.	CLO2	[3]
	(d)	Determine the force between two free electrons spaced 1Å apart.	CLO3	[3]
Q2.	**(a)**	Define the terms: (i) Electric Field and (ii) Electric Potential	CLO1	[2]
	(b)	Deduce electric field strength for a point charge.	CLO2	[4]
	(c)	Two point charges q_1 and q_2 are 3m apart, and their combined charge is 20 μC. what are the magnitude of the two charges if (a) one repels the other with a force of 0.075N and (b) one attracts the other with a force of 0.525N.	CLO3	[4]
Q3.	**(a)**	Define the terms: (i) line charge density (ii) surface charge density (iii) space charge density	CLO1	[2]
	(b)	Deduce electric field strength on the axis of a ring of radius R.	CLO2	[4]
	(c)	Find the electric field for a uniform charged wire at a point P a perpendicular distance from the wire.	CLO2	[4]
Q4.	**(a)**	Define and explain electric dipole moment with figure.	CLO1	[2]
	(b)	Find an expression for the potential at a point due to a dipole.	CLO2	[4]
	(c)	An electric dipole consists of two opposite charges of magnitude 2×10^{-6} C separated by a distance 1.0 cm. It is placed in an external electric field of 2.0×10^5 N/C. (a) What maximum torque does the field exert on the dipole? (b) How much work must an external agent do to turn the dipole from its initial alignment given by $\theta = 0°$ to final alignment $\theta = 90°$.	CLO3	[4]
Q5.	**(a)**	State and Explain Gauss's law for electrostatics.	CLO2	[2]
	(b)	Apply Gauss's law for a cylinder and find out the total flux φ_E	CLO4	[4]
	(c)	Find an expression for the potential at a point in an electric field due to a point charge.	CLO2	[4]

HAMDARD UNIVERSITY BANGLADESH

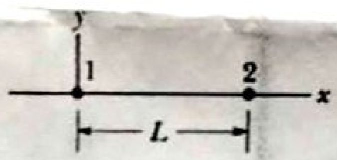
Department of Electrical and Electronics Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY 111 — **Course Title: Physics II Electricity and Magnetism**
Mid-Term Examination — Semester: Spring 2026
Date: 6/22/2026 — Total Time: 1:30 Hours — **Total Marks: 30**

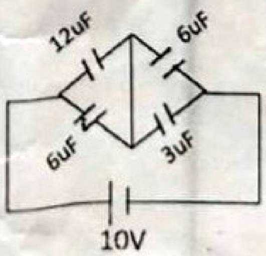
Answer any three (03) out of the following questions: $3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

			CLOs	Marks
Q1.	**(a)**	Explain why electric charge is quantized.	CLO1	[2]
	(b)	State and explain gauss's law	CLO1	[3]
	(c)	Find the electric field due to an infinitely long straight wire.	CLO3	[3]
	(d)	A straight wire of length 100 m has a linear charge density of 1 μC/m. Find the electric field at a point 0.5 m from the wire.	CLO3	[2]
Q2.	**(a)**	State coulomb's law.	CLO1	[2]
	(b)	Find the electric potential due to a uniformly charged rod at an axial point.	CLO3	[5]
	(c)	In Fig. , particle 1 of charge +6.0 μC and particle 2 of charge −2.0 μC are held at separation L = 10.0 cm on an x axis. If particle 3 of unknown charge q3 is to be located such that the net electrostatic force on it from particles 1 and 2 is zero, what must be the (a) x and (b) y coordinates of particle 3?	CLO3	[3]



			CLOs	Marks
Q3.	**(a)**	What is an electric dipole moment?	CLO1	[2]
	(b)	Define electric potential and electric field. Hence, establish the relationship between them.	CLO2	[3]
	(c)	For an electric dipole having charges +q and −q separated by a distance l, with O as the midpoint, Find the electric potential at any external point P located at a distance r from the center and making an angle θ with the dipole axis.	CLO3	[5]
Q4.	**(a)**	Define capacitance?	CLO1	[2]
	(b)	(i) Find the Equivalent Capacitance of the circuit. (ii) Find charge storage on 3uF and 12 uF Capacitor.	CLO4	[8]



Q5. (a) What is the Hall effect? CLO2 [2]

(b) Explain the Direction of induced current with net sketch. CLO2 [3]

(c) Deduce the mathematical expression for Hall voltage, $V_H = \frac{BI}{pw}$ CLO3 [5]